Precalculus II

An Original Ten-Week Course Reader

Dr. Lee MATH& 142

Contents

1 Angles, Rotation, and the Unit Circle **4**
- 1.1 Angles measure turning . . . 4
- 1.2 Radian measure . . . 4
- 1.3 Arc length and angular speed . . . 5
- 1.4 Coordinates on the unit circle . . . 6
- 1.5 Reference angles and periodicity . . . 7
- 1.6 Where exact coordinates come from . . . 7
- 1.7 From the circle to a wave . . . 7
- 1.8 Example: a Ferris-wheel model . . . 8
- 1.9 Exercises . . . 8

2 The Six Trigonometric Functions **10**
- 2.1 From a point to six functions . . . 10
- 2.2 Tangent as a slope . . . 10
- 2.3 Signs and the quadrants . . . 11
- 2.4 Domains and undefined values . . . 12
- 2.5 Zeros and vertical asymptotes . . . 12
- 2.6 Right-triangle ratios . . . 12
- 2.7 Exact values at special angles . . . 13
- 2.8 Graphs of the trigonometric functions . . . 13
- 2.9 Applications: orientation and measurement . . . 14
- 2.10 Measurement with trigonometric ratios . . . 14
- 2.11 Exercises . . . 14

3 Periodic Functions and Their Graphs **16**
- 3.1 Circular motion and periodicity . . . 16
- 3.2 Amplitude and the midline . . . 16
- 3.3 Vertical transformations . . . 16
- 3.4 Period and horizontal scaling . . . 17
- 3.5 Phase shift . . . 17
- 3.6 Five landmark points . . . 17
- 3.7 Models of repeating motion . . . 18
- 3.8 Constructing a model from measurements . . . 18
- 3.9 Tangent and reciprocal graphs . . . 18
- 3.10 Exercises . . . 19

4 Inverse Trigonometry and Identities **20**
4.1 Inverse functions and principal branches . . . 20
4.2 Reflection across the diagonal . . . 20
4.3 Compositions and triangle geometry . . . 21
4.4 Identities and their domains . . . 22
4.5 Symmetry identities . . . 22
4.6 Angle addition and double angles . . . 22
4.7 A geometric derivation of the addition formulas . . . 22
4.8 Double-angle and half-angle consequences . . . 23
4.9 Applications: measured angles . . . 23
4.10 Recovering angles in measurement problems . . . 23
4.11 Exercises . . . 23

5 Trigonometric Equations and Applications **25**
5.1 A first equation on the unit circle . . . 25
5.2 Equations with transformations . . . 25
5.3 Factoring and checking domains . . . 25
5.4 Trigonometric identities in equations . . . 25
5.5 Solving triangles: notation . . . 26
5.6 The ambiguous SSA case . . . 26
5.7 Applied modeling and interpretation . . . 26
5.8 Exercises . . . 26

6 Vectors and Geometric Applications **28**
6.1 From coordinates to directed displacements . . . 28
6.2 Vector arithmetic . . . 28
6.3 Direction and trigonometry . . . 28
6.4 The dot product measures alignment . . . 29
6.5 Projection and useful components . . . 29
6.6 A geometric connection to the Law of Cosines . . . 29
6.7 Exercises . . . 29

7 Polar Coordinates and Complex Numbers **31**
7.1 Two descriptions of the same point . . . 31
7.2 Polar coordinates are not unique . . . 31
7.3 Complex numbers as plane vectors . . . 31
7.4 Addition and multiplication mean different geometries . . . 32
7.5 De Moivre's theorem and roots . . . 32
7.6 Polar graphs: a different way to draw a curve . . . 32
7.7 Exercises . . . 32

8 Parametric Equations and Conic Sections **34**
8.1 A moving point traces a curve . . . 34
8.2 Eliminating the parameter . . . 34
8.3 Projectile motion as a parametric curve . . . 34
8.4 Conics: circles and ellipses . . . 35
8.5 Parabolas and their focus . . . 35
8.6 Hyperbolas and differences of distances . . . 35
8.7 Changing descriptions . . . 35

8.8 Exercises 35

9 Sequences and Series **37**
9.1 A sequence is an ordered list 37
9.2 Arithmetic sequences: repeated addition 37
9.3 Geometric sequences: repeated multiplication 37
9.4 Sigma notation 38
9.5 Infinite geometric series 38
9.6 Recursive rules beyond arithmetic and geometric patterns 38
9.7 Exercises 38

10 Synthesis: Motion and Patterns **39**
10.1 One motion, several descriptions 39
10.2 Velocity, vectors, and periodicity 39
10.3 Sampling creates a sequence 39
10.4 Scaling and rotation together 40
10.5 Sums represent accumulated effects 40
10.6 Different curves arise from different constraints 40
10.7 A complete example 40
10.8 Exercises 40

1 Angles, Rotation, and the Unit Circle

Angles provide a description of rotation. On the unit circle, this description determines a point whose coordinates define the sine and cosine functions.

A rotation can be specified by its angle, independently of the size of the circle on which it acts. The relation between angle and position leads to trigonometric coordinates and to models of periodic motion.

Learning objectives. The principal objectives are to describe signed rotation, convert between degrees and radians, calculate arc length and angular speed, determine exact unit-circle coordinates, and relate uniform circular motion to periodic coordinate functions.

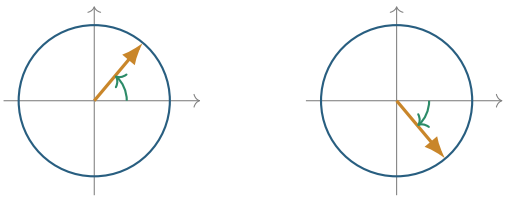
1.1 Angles measure turning

An angle records a rotation from one ray to another. Counterclockwise rotations are positive, and clockwise rotations are negative. A rotation can exceed one full revolution. For example, 450° and 90° have the same terminal ray.

Definition 1.1. Two angles are *coterminal* if they have the same initial and terminal rays. They differ by an integer multiple of 360°, or 2π radians.

Example 1.1. A positive coterminal angle for -50° is $-50^\circ + 360^\circ = 310^\circ$.

In standard position, the initial ray coincides with the positive horizontal axis. Positive angles correspond to counterclockwise rotation and negative angles to clockwise rotation. The signed angle specifies the accumulated rotation, whereas the terminal ray specifies only the final direction. Thus 30°, 390°, and -330° point along the same terminal ray, although their rotations are different.



Worked example: multiple turns. Find an angle between 0° and 360° coterminal with -770°. Adding 360° twice gives -50°; adding it once more gives 310°. The resulting angle lies in $[0^\circ, 360^\circ)$ and has the same terminal ray.

Try it. Sketch -120°, 240°, and 600°. Which terminal rays coincide? Justify the result using coterminal angles.

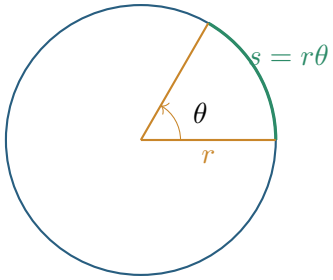
1.2 Radian measure

For a circle of radius r and an arc of length s subtended by a central angle, the magnitude of the angle in radians is s/r. For an oriented arc, the signed arc length determines the signed angle. A full circumference is $2\pi r$, so one revolution measures 2π radians:

$$\boxed{\theta = \frac{s}{r}}, \qquad 180^\circ = \pi \text{ radians}.$$

Example 1.2. Convert 150° to radians: $150(\pi/180) = 5\pi/6$. Convert $7\pi/4$ radians to degrees: $(7\pi/4)(180/\pi) = 315^\circ$.

Try it. Convert 225° to radians and $-\pi/3$ radians to degrees, and sketch their terminal rays.



Arc length and central angle

For a fixed central angle, the intercepted arc length is proportional to the radius. Consequently, the ratio s/r does not depend on the size of the circle.

Degrees	Radians	Fraction of one turn
30°	$\pi/6$	$1/12$
45°	$\pi/4$	$1/8$
60°	$\pi/3$	$1/6$
90°	$\pi/2$	$1/4$
180°	π	$1/2$
270°	$3\pi/2$	$3/4$
360°	2π	1

The conversion $180^\circ = \pi$ radians determines every entry in the table. In particular, $60^\circ = \pi/3$ radians.

Example 1.3. Convert a negative angle: $-210^\circ \cdot \pi/180^\circ = -7\pi/6$. Its terminal ray coincides with $5\pi/6$, because adding 2π to $-7\pi/6$ gives $5\pi/6$.

1.3 Arc length and angular speed

For signed radian measure θ, the directed arc length is $s = r\theta$; the corresponding distance traveled is $r|\theta|$. Over a time interval of length t, the average angular velocity is $\omega = \theta/t$ and the average tangential speed is $v = r|\omega|$. For uniform rotation, these quantities are constant.

Example 1.4. A wheel of radius 0.4 m turns through $3\pi/2$ radians. A point on its rim travels $s = 0.4(3\pi/2) = 0.6\pi$ m.

Try it. A Ferris wheel of radius 12 m makes one revolution every 40 s. Find its angular speed and a passenger's tangential speed.

Worked example: a carousel. A carousel makes one revolution every 24 seconds. A child sits 3 meters from the axis, while another sits 5 meters from the axis. Both riders have the same angular speed,

$$\omega = \frac{2\pi}{24} = \frac{\pi}{12} \text{ rad/s}.$$

Their tangential speeds differ:

$$v_1 = 3\frac{\pi}{12} = \frac{\pi}{4} \text{ m/s}, \qquad v_2 = 5\frac{\pi}{12} = \frac{5\pi}{12} \text{ m/s}.$$

Tangential speed depends on the distance from the axis, whereas angular speed does not.

The arc-length formula requires radian measure. For example, a quarter-turn is $\pi/2$ radians, so its arc length is $\pi r/2$.

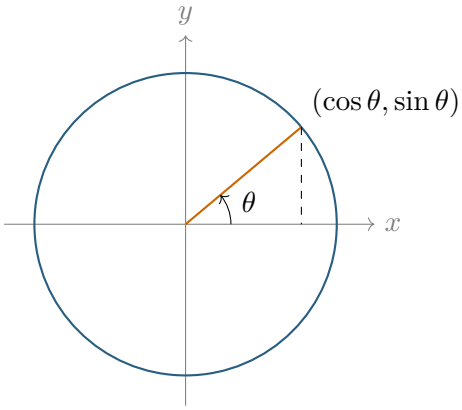
Try it. A wheel of radius 0.6 m turns through 135° in 3 seconds. Find the arc length, average angular speed, and average tangential speed. Explain the units of all three answers.

1.4 Coordinates on the unit circle

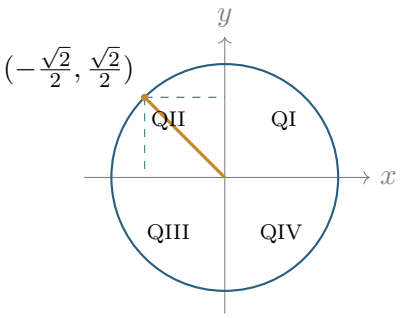
Take a circle of radius 1 centered at the origin. The point at angle θ, measured from the positive x-axis, has coordinates

$$\boxed{(x, y) = (\cos\theta, \sin\theta)}.$$

Cosine is the horizontal coordinate and sine is the vertical coordinate. Because the point lies on $x^2 + y^2 = 1$, we get $\cos^2\theta + \sin^2\theta = 1$.



Rotating the point $(1, 0)$ through θ produces $(\cos\theta, \sin\theta)$. Both coordinates are signed quantities; their signs depend on the quadrant containing the terminal ray.



Worked example: a point in Quadrant III. At $225° = 5\pi/4$, the reference angle is $\pi/4$. The two coordinates have equal absolute value, but both are negative:

$$(\cos(5\pi/4), \sin(5\pi/4)) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right).$$

The reference triangle determines the coordinate magnitudes; the quadrant determines their signs.

Try it. Without a calculator, describe the point on the unit circle at 150° and at 330°. Which coordinates have the same absolute value? Which signs differ?

1.5 Reference angles and periodicity

Special angles arise from right triangles:

θ	$\cos\theta$	$\sin\theta$
0	1	0
$\pi/6$	$\sqrt{3}/2$	$1/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3$	$1/2$	$\sqrt{3}/2$
$\pi/2$	0	1

The reference angle supplies the absolute values in every quadrant; the quadrant gives the signs.

Example 1.5. At $5\pi/6$, the reference angle is $\pi/6$ and the point is in quadrant II, so $\cos(5\pi/6) = -\sqrt{3}/2$ and $\sin(5\pi/6) = 1/2$.

After one full revolution the point returns to its previous position:

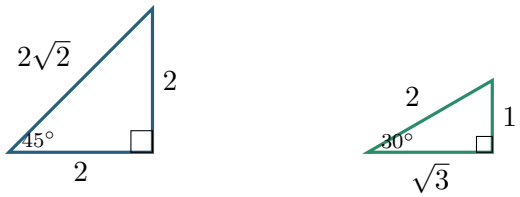
$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

This geometric periodicity will become the foundation of sinusoidal graphs.

1.6 Where exact coordinates come from

The standard exact values follow from the geometry of two special right triangles.

An isosceles right triangle with legs 1 has hypotenuse $\sqrt{2}$. Dividing both legs by $\sqrt{2}$ yields the point $(\sqrt{2}/2, \sqrt{2}/2)$ at 45°. Likewise, a 30°–60°–90° triangle has side lengths in the ratio $1 : \sqrt{3} : 2$. Dividing by its hypotenuse produces the coordinates $(\sqrt{3}/2, 1/2)$ for 30° and $(1/2, \sqrt{3}/2)$ for 60°.



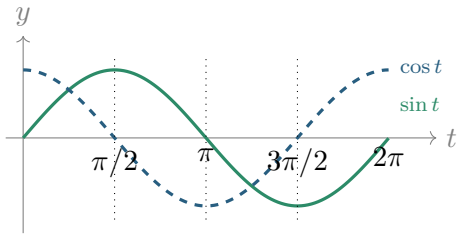
On the unit circle, the hypotenuse is normalized to length 1, and the remaining side lengths give the absolute coordinate values.

1.7 From the circle to a wave

For uniform rotation with angular velocity ω and initial angle zero, let

$$\theta(t) = \omega t, \qquad x(t) = \cos(\omega t), \quad y(t) = \sin(\omega t).$$

The point traces a circle in the coordinate plane; each coordinate, regarded as a function of time, oscillates periodically.



These graphs represent coordinate values as functions of time, rather than the circular trajectory in the xy-plane.

1.8 Example: a Ferris-wheel model

A Ferris wheel has radius 8 meters, its center is 10 meters above ground, and a passenger starts at the rightmost point. Suppose it rotates counterclockwise once every 40 seconds. Its angular speed is $\omega = 2\pi/40 = \pi/20$ radians per second, and the passenger's position relative to the ground is

$$\boxed{x(t) = 8\cos\left(\frac{\pi t}{20}\right), \qquad y(t) = 10 + 8\sin\left(\frac{\pi t}{20}\right).}$$

At $t = 0$ the passenger is at $(8, 10)$. At $t = 10$ seconds the wheel has turned through $\pi/2$, so the passenger is at $(0, 18)$. At $t = 20$ seconds the passenger is at $(-8, 10)$. The coordinate equations determine the passenger's position at any time.

The radius determines the amplitude of both coordinate functions. The center height shifts the vertical coordinate, and the period determines the angular velocity.

Try it. A different wheel has radius 6 meters, center height 7 meters, and period 30 seconds. Starting from the rightmost point, write its coordinate model and find the passenger's height after 7.5 seconds.

1.9 Exercises

1. Convert $-135°$ and $300°$ to radians.
2. Convert $11\pi/6$ and $-5\pi/4$ to degrees.
3. Find an angle in $[0, 2\pi)$ coterminal with $19\pi/6$.
4. A circle of radius 8 cm subtends an angle of $5\pi/12$. Find the arc length.
5. A turntable makes $33\frac{1}{3}$ revolutions per minute. Find angular speed in radians per second.
6. Explain why sine and cosine must lie between -1 and 1.
7. Find unit-circle coordinates at $2\pi/3$, $5\pi/4$, and $11\pi/6$.
8. A Ferris wheel has radius 10 m and center $(0, 12)$. Write a rider's coordinates after turning through θ from the rightmost point.
9. A point begins at $(1, 0)$ and turns through $-7\pi/6$ radians. Determine a coterminal angle in $[0, 2\pi)$ and give the exact point.
10. A wheel completes 9 revolutions in 2 minutes. Determine its angular speed in both radians per second and degrees per second.

11. A rider on a circle of radius 4 m moves through 75°. Find the arc length exactly in terms of π and approximately to two decimals.

12. Describe the coordinates at $3\pi/2$ and explain why a terminal point on an axis does not belong to any open quadrant.

13. For the Ferris-wheel model in the chapter, explain which physical quantity would change if the coefficient 8 in both coordinate equations became 12.

The unit-circle coordinates also determine tangent, cotangent, secant, and cosecant, which are developed in the following chapter.

2 The Six Trigonometric Functions

The six trigonometric functions are defined by coordinates on the unit circle and ratios of those coordinates. Their relationships determine their signs, domains, and graphs.

Let $P = (x, y)$ be the point at angle θ on the unit circle. Its coordinates are $x = \cos\theta$ and $y = \sin\theta$. The remaining four functions are formed from quotients and reciprocals of these coordinates.
Learning objectives. Define the six functions on the unit circle, calculate exact values in each quadrant, determine their domains, identify characteristic graph features, and apply trigonometric ratios to measurement problems.

2.1 From a point to six functions

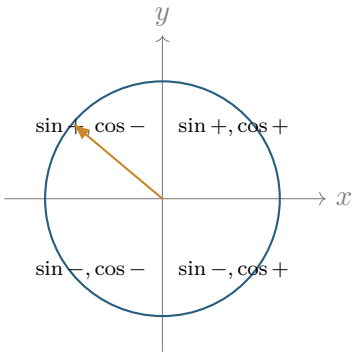
Last week we studied the unit-circle point $P = (x, y) = (\cos\theta, \sin\theta)$. The two coordinates determine four additional ratios whenever the denominators are nonzero:

$$\sin\theta = y, \quad \cos\theta = x, \quad \tan\theta = \frac{y}{x},$$
$$\csc\theta = \frac{1}{y}, \quad \sec\theta = \frac{1}{x}, \quad \cot\theta = \frac{x}{y}.$$

Thus the six functions encode related properties of a single point.

Definition 2.1. The *tangent* function is $\tan\theta = \sin\theta/\cos\theta$ where $\cos\theta \neq 0$. The functions csc, sec, cot are the reciprocals of sin, cos, tan respectively, wherever defined.

Example 2.1. At $\theta = \pi/3$ the circle point is $(1/2, \sqrt{3}/2)$. Thus $\tan\theta = \sqrt{3}$, $\sec\theta = 2$, $\csc\theta = 2/\sqrt{3} = 2\sqrt{3}/3$, and $\cot\theta = 1/\sqrt{3} = \sqrt{3}/3$.

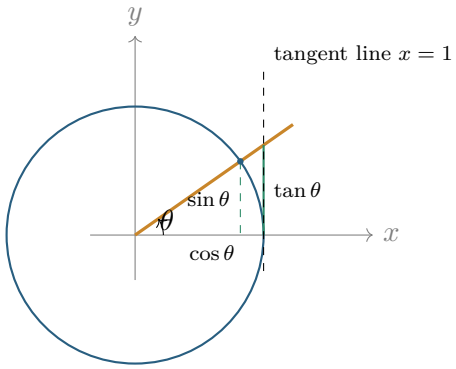


2.2 Tangent as a slope

For $\cos\theta \neq 0$, the line through the origin and $P = (\cos\theta, \sin\theta)$ has slope

$$m = \frac{\text{rise}}{\text{run}} = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

Tangent therefore gives the slope of the terminal line. Its magnitude increases without bound as the line approaches a vertical position, at which the slope is undefined.



The terminal line intersects $x = 1$ at $(1, \tan\theta)$, whenever $\cos\theta \neq 0$. Hence tangent also has a geometric interpretation as a directed length on a line tangent to the unit circle.

Example 2.2. At $30°$, the unit-circle coordinates are $(\sqrt{3}/2, 1/2)$. The slope of the corresponding ray is

$$\tan 30° = \frac{1/2}{\sqrt{3}/2} = \frac{\sqrt{3}}{3}.$$

This is also the slope of a line rising one unit for every $\sqrt{3}$ units of horizontal displacement.

2.3 Signs and the quadrants

The coordinate signs determine every trig-function sign. In quadrant I both coordinates are positive; in II, only sine is positive; in III, tangent is positive because both coordinates are negative; and in IV, only cosine is positive. Reciprocal functions have the same signs as their corresponding original functions.

Example 2.3. For $\theta = 5\pi/4$, the reference angle is $\pi/4$ and $P = (-\sqrt{2}/2, -\sqrt{2}/2)$. Therefore $\tan\theta = 1$, $\sec\theta = -\sqrt{2}$, and $\csc\theta = -\sqrt{2}$.

For a useful sign check, consider $150° = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$. Both sine and cosecant are positive, both cosine and secant are negative, and tangent and cotangent are negative:

$$\tan(5\pi/6) = -\frac{\sqrt{3}}{3}, \qquad \sec(5\pi/6) = -\frac{2\sqrt{3}}{3}, \qquad \csc(5\pi/6) = 2.$$

The signs of the six functions follow directly from the signs of sine and cosine.

	sin	cos	tan	csc	sec	cot
Quadrant I	+	+	+	+	+	+
Quadrant II	+	−	−	+	−	−
Quadrant III	−	−	+	−	−	+
Quadrant IV	−	+	−	−	+	−

Try it. The terminal point of an angle lies in quadrant IV and satisfies $\cos\theta = 3/5$. Determine $\sin\theta$, $\tan\theta$, and their reciprocals, showing how the quadrant tells you the signs.

2.4 Domains and undefined values

Division by zero is undefined. In particular, $\tan\theta$ and $\sec\theta$ do not exist where $\cos\theta = 0$, at $\pi/2 + k\pi$. Similarly $\cot\theta$ and $\csc\theta$ do not exist where $\sin\theta = 0$, at $k\pi$, for any integer k. Undefined values must be distinguished from zeros of a function.

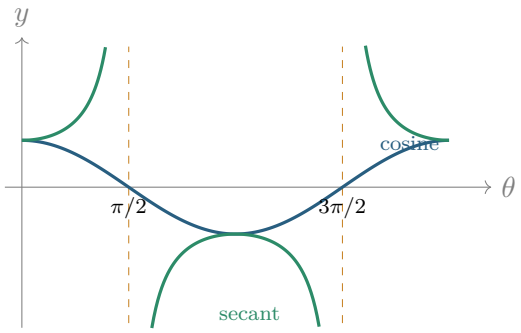
Try it. Find all six trig-function values at $\theta = 3\pi/2$, identifying every undefined value.

2.5 Zeros and vertical asymptotes

Near zeros of their denominators, trigonometric quotients may become arbitrarily large in magnitude. This behavior accounts for their vertical asymptotes.

For example, as θ approaches $\pi/2$ from the left, $\cos\theta$ approaches zero from above. Consequently $\sec\theta = 1/\cos\theta$ increases without bound. From the right, cosine is slightly negative, and secant becomes a large negative number. At $\pi/2$ itself, secant has no value.

The line $\theta = \pi/2$ is a vertical asymptote of both tangent and secant.



Only those portions of the secant graph within the displayed vertical range are shown. Its branches are separated by vertical asymptotes.

Try it. Describe the same phenomenon for $\csc\theta$ near $\theta = 0$. Explain why a zero of sine becomes an undefined value of cosecant.

2.6 Right-triangle ratios

For an acute angle θ in a right triangle, scale the triangle so its hypotenuse has length 1. The opposite and adjacent sides then give the coordinates $\sin\theta$ and $\cos\theta$. Restoring a general hypotenuse h gives

$$\sin\theta = \frac{\text{opposite}}{h}, \quad \cos\theta = \frac{\text{adjacent}}{h}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The unit-circle definitions extend the right-triangle ratios to arbitrary real angles.

Example 2.4. A ramp rises 1.5 meters over a horizontal run of 6 meters. Its angle of inclination satisfies $\tan\theta = 1.5/6 = 1/4$. Therefore $\theta = \arctan(1/4) \approx 14.0°$.

Writing o, a, and h for opposite side, adjacent side, and hypotenuse, respectively, gives

	opposite and hypotenuse	adjacent and hypotenuse	opposite and adjacent
direct	$\sin\theta = o/h$	$\cos\theta = a/h$	$\tan\theta = o/a$
reciprocal	$\csc\theta = h/o$	$\sec\theta = h/a$	$\cot\theta = a/o$

The six ratios describe the same triangle and agree with the unit-circle definitions for acute angles.

Example 2.5. A support cable stretches 13 meters from the ground to a point 12 meters above it. The horizontal run is $\sqrt{13^2 - 12^2} = 5$ meters. If θ is the angle with the ground, then

$$\sin\theta = \frac{12}{13}, \quad \cos\theta = \frac{5}{13}, \quad \tan\theta = \frac{12}{5}, \quad \csc\theta = \frac{13}{12}, \quad \sec\theta = \frac{13}{5}, \quad \cot\theta = \frac{5}{12}.$$

Each quotient expresses a relation between lengths of the same triangle.

2.7 Exact values at special angles

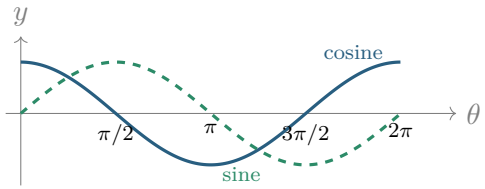
An equilateral triangle of side 2 splits into two right triangles with sides $1, \sqrt{3}, 2$. This produces $\sin 30° = 1/2$ and $\cos 30° = \sqrt{3}/2$. A square of side 1 has diagonal $\sqrt{2}$, producing $\sin 45° = \cos 45° = \sqrt{2}/2$.



2.8 Graphs of the trigonometric functions

On a trigonometric graph, the independent variable is an angle and the dependent variable is a coordinate or ratio at that angle. Such a graph is distinct from the circular trajectory of the associated point.

Sine and cosine remain between -1 and 1. Both repeat after a full rotation, so their fundamental period is 2π radians, or 360°. Tangent and cotangent repeat after half a turn, so their fundamental period is π radians, or 180°. Secant and cosecant share the 2π period of their reciprocal partners.



The zeros, excluded inputs, and fundamental period determine important features of a trigonometric graph.

Example 2.6. Rotation by π changes the signs of both unit-circle coordinates. Thus

$$\frac{\sin(\theta + \pi)}{\cos(\theta + \pi)} = \frac{-\sin\theta}{-\cos\theta} = \tan\theta.$$

Therefore tangent has period π on its domain.

The accompanying Six Trigonometric Graphs and Reciprocal Trigonometric Graphs demonstrations illustrate these functions and their asymptotes.

Try it. Sketch $y = \tan x$ and $y = \cot x$ on $(0, \pi)$, marking their zeros and asymptotes. How do the two plots differ at the interval's endpoints?

2.9 Applications: orientation and measurement

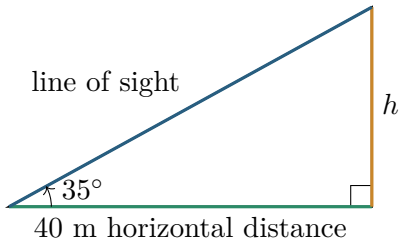
Suppose an observer stands 40 m from the foot of a vertical tower. The angle of elevation to its top is 35°. The tower's height h satisfies

$$\tan 35^\circ = \frac{h}{40}, \qquad h = 40 \tan 35^\circ \approx 28.0 \text{ m}.$$

Here the tangent function links an angle to a ratio of lengths. Always sketch the physical situation and label which distances are opposite and adjacent before calculating.

2.10 Measurement with trigonometric ratios

The appropriate trigonometric ratio depends on the available lengths. Tangent relates height to horizontal distance; sine relates height to line-of-sight distance.



Worked example: height measurement. An observer is 20 meters from the base of a tree and measures an angle of elevation of 40° to its top. If the observer's eyes are 1.6 meters above ground, the elevation of the treetop above eye level is

$$h_{\text{above eyes}} = 20 \tan 40^\circ \approx 16.8 \text{ m}.$$

The tree height is therefore about $16.8 + 1.6 = 18.4$ meters. The observer's eye height must be added to determine the height above ground.

Try it. A rescue drone is observed at an elevation angle of 25° from an observer whose eye level is 1.7 meters above ground. The point directly beneath the drone is 90 meters away horizontally. Estimate the drone's altitude.

2.11 Exercises

1. At $\theta = 2\pi/3$, find all six functions exactly.
2. At $\theta = 7\pi/4$, find $\tan, \cot, \sec, \csc$.
3. Explain geometrically why $\tan(\pi/2)$ is undefined.
4. For which angles in $[0, 2\pi)$ is $\csc\theta$ undefined?
5. A right triangle has opposite side 5 and adjacent side 12. Find all six trig ratios exactly.
6. A ladder leans against a wall, reaching 8 m high. Its foot is 3 m from the wall. Find the angle with the ground.
7. A drone is viewed at an elevation angle of 28° from a point 75 m horizontally from the point below it. Estimate its altitude.

8. Show that $\sin^2\theta + \cos^2\theta = 1$ implies $1 + \tan^2\theta = \sec^2\theta$ wherever $\cos\theta \neq 0$.
9. Find the exact values of all six trig functions at $11\pi/6$.
10. Describe the zeros and vertical asymptotes of $y = \tan x$ on $[-2\pi, 2\pi]$.
11. Sketch one period of $y = \sec x$, labeling asymptotes and its lowest positive and highest negative values.
12. Explain why neither $\csc x$ nor $\sec x$ can equal $1/2$.
13. If $\cos\theta = -12/13$ and θ lies in quadrant II, determine all other trig values exactly.
14. Show using the unit circle that $\tan(\theta + \pi) = \tan\theta$ wherever both sides are defined.
15. A point on a rotating wheel has unit-circle coordinates $(0, -1)$. Identify the angles in $[0, 2\pi)$ and all four undefined or zero-valued functions.

3 Periodic Functions and Their Graphs

Periodic functions describe quantities that repeat as their input increases. Trigonometric functions provide fundamental examples: their periodicity arises from rotation, and their graphs can be transformed to represent a wide range of oscillatory processes.

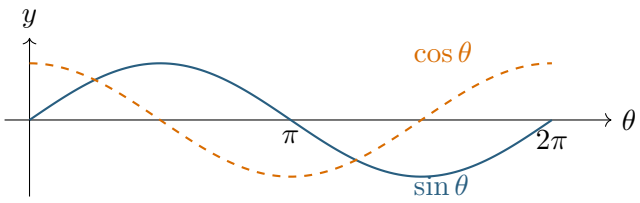
Learning objectives. Identify periods, amplitudes, midlines, and phase shifts; construct transformed sine and cosine graphs; distinguish angular frequency from frequency; and determine sinusoidal models from geometric or measured information.

3.1 Circular motion and periodicity

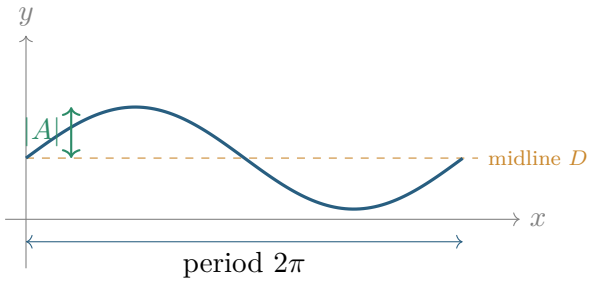
For a point traveling around the unit circle with angle θ, its vertical coordinate is $y = \sin\theta$ and its horizontal coordinate is $x = \cos\theta$. Each revolution restores the same coordinates. Hence

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

A function f is *periodic* if there exists $T > 0$ such that $f(x + T) = f(x)$ for every x in its domain. When a least positive such T exists, it is the *fundamental period*. The fundamental periods of sine and cosine are both 2π. Moreover, $\sin 0 = 0$ and $\cos 0 = 1$.



3.2 Amplitude and the midline



3.3 Vertical transformations

Consider $y = A\sin x + D$. Its midline is $y = D$, its amplitude is $|A|$, and its range is $[D - |A|, D + |A|]$. A negative A reflects the wave across the midline.

Example 3.1. The function $y = 3\sin x - 2$ has amplitude 3, midline $y = -2$, maximum 1, and minimum -5.

For a sinusoid with maximum M and minimum m, the amplitude and midline are

$$|A| = \frac{M - m}{2}, \qquad D = \frac{M + m}{2}.$$

These relations determine the vertical parameters without knowing the phase or period.

Example 3.2. A periodically varying temperature has maximum 26°C and minimum 14°C. Its midline is 20°C and its amplitude is 6°C. These values do not determine the period or phase.

Try it. A sinusoidal signal ranges from -7 to 5. Determine its amplitude, midline, and range, and sketch one possible cycle.

3.4 Period and horizontal scaling

For $y = \sin(Bx)$ with $B \neq 0$, one full cycle occurs when Bx changes by 2π; the period is $T = 2\pi/|B|$.

Example 3.3. The graph of $y = \cos(4x)$ repeats every $\pi/2$ units of input.

When x denotes time measured in seconds, the quantity $|B|$ is angular frequency in radians per second, and ordinary frequency is $f = 1/T = |B|/(2\pi)$ cycles per second.

A phase change of 2π radians corresponds to one cycle. If the argument of a sinusoidal function is $\omega t + \phi$, then its period is $T = 2\pi/|\omega|$, and its frequency is

$$f = \frac{1}{T} = \frac{|\omega|}{2\pi}.$$

Frequency is measured in cycles per unit time, while angular frequency is measured in radians per unit time. These units describe the same repetition rate in different ways.

Example 3.4. A mechanism rotates at 90 revolutions per minute. Its frequency is 1.5 cycles per second, and its angular speed is $2\pi(1.5) = 3\pi$ radians per second. Its period is 2/3 second.

3.5 Phase shift

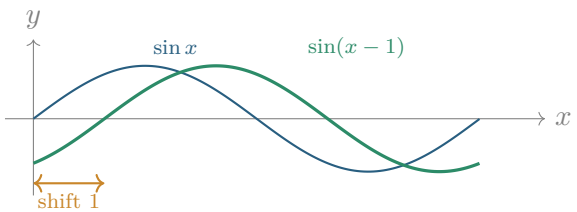
A useful general form is

$$\boxed{y = A\sin(B(x - C)) + D}.$$

Here C is the horizontal shift, D the midline, $|A|$ the amplitude, and $2\pi/|B|$ the period. We should distinguish $B(x - C)$ from $Bx - C$: factoring first prevents an incorrect phase shift.

Example 3.5. For $y = 2\sin(3x - \pi) + 1 = 2\sin(3(x - \pi/3)) + 1$, the period is $2\pi/3$, the shift is $\pi/3$ to the right, the amplitude is 2, and the midline is 1.

The parameterization assumes $B \neq 0$. In factored form, the phase shift is C. In the equivalent form $A\sin(Bx + \phi) + D$, the same shift is $C = -\phi/B$. The sign of B may be taken positive in a standard sine model by modifying the sign of A.



3.6 Five landmark points

Rather than sketching a sinusoid by guesswork, divide one period into four equal intervals. For $A > 0$, sine begins on the midline, rises to a maximum after one quarter-period, returns to the midline, reaches a minimum, then returns to the midline. A negative amplitude reverses the rises and falls.

Try it. Find five landmark points of $y = -2\cos(\pi(x-1)) + 3$, showing one entire period.

For $y = A\sin(B(x-C)) + D$ with $B > 0$, write $T = 2\pi/B$. One cycle is determined by the following five points:

x	C	$C+T/4$	$C+T/2$	$C+3T/4$	$C+T$
y	D	$D+A$	D	$D-A$	D

For cosine, the corresponding outputs are $D+A, D, D-A, D, D+A$. These points determine the positions of maxima, minima, and midline crossings; the smooth sinusoidal curve passes through them.

Example 3.6. For $y = -2\cos(\pi(x-1)) + 3$, the period is $T = 2$. Beginning at $x = 1$, the five landmark points are

$$(1,1), \quad (3/2,3), \quad (2,5), \quad (5/2,3), \quad (3,1).$$

The negative coefficient places the minimum at the start of the cycle.

3.7 Models of repeating motion

A Ferris wheel has radius 9 m, its center is 11 m above ground, and it makes one revolution every 30 s. If a passenger begins at the rightmost position and travels counterclockwise, their height is

$$h(t) = 11 + 9\sin\left(\frac{2\pi}{30}t\right).$$

Its amplitude is 9 m, midline is 11 m, and period is 30 s. A sinusoidal model can describe a wheel, a spring, a tide, or a musical tone, provided its parameters actually correspond to the chosen physical quantities.

3.8 Constructing a model from measurements

Suppose a buoy reaches a maximum height of 5 meters and a minimum height of 1 meter. Successive maxima are separated by 8 seconds. Taking $t = 0$ at a maximum yields the model

$$h(t) = 3 + 2\cos\left(\frac{\pi t}{4}\right).$$

The midline is 3 meters, the amplitude is 2 meters, and the angular frequency is $\pi/4$ radians per second. If the time origin were chosen at an upward midline crossing, a sine model would be more convenient. The physical motion is unchanged by this choice of representation.

Try it. A tide measurement varies between 0.8 m and 3.6 m and repeats every 12 hours. Write a cosine model assuming that the maximum occurs at $t = 2$ hours. State the meaning and units of every parameter.

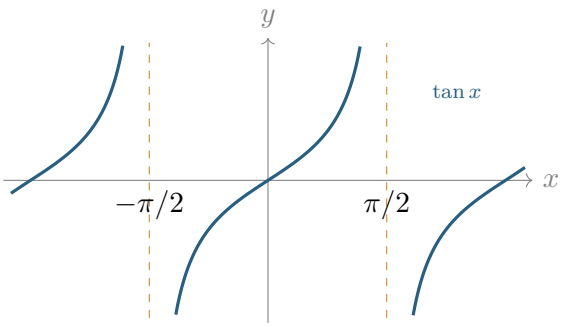
3.9 Tangent and reciprocal graphs

The tangent function repeats after π, not 2π, because both sine and cosine change sign after a half turn and their ratio remains the same:

$$\tan(x+\pi) = \tan x.$$

Tangent has vertical asymptotes at $x = \pi/2 + k\pi$. The secant and cosecant graphs arise as reciprocals of cosine and sine; their breaks occur wherever the respective denominator is zero.

Because $\tan x = \sin x/\cos x$, its zeros occur at $x = k\pi$ and its vertical asymptotes at $x = \pi/2 + k\pi$. Similarly, $\cot x = \cos x/\sin x$ is undefined at integer multiples of π. The graphs of secant and cosecant lie outside the strip $-1 < y < 1$ because their absolute values are at least 1 where defined.



3.10 Exercises

1. Find the amplitude, midline, period, and range of $y = -4\sin(2x) + 1$.
2. Rewrite $y = 3\cos(5x - \pi) + 2$ in shifted form and identify the phase shift.
3. Give five landmark points for $y = 2\sin(x - \pi/2)$.
4. Explain why $y = \sin(2x)$ completes two periods on $[0, 2\pi]$.
5. A wheel of radius 6 m rotates once every 24 seconds. Its center is 8 m above the ground. Model the rider's height when starting at the rightmost point.
6. Identify the vertical asymptotes of $y = \tan(2x)$ on $[0, 2\pi)$.
7. Sketch one period of $y = 2\sec x$ and indicate all discontinuities.
8. Explain how you can determine amplitude and midline directly from observed maximum and minimum values.
9. Determine a sinusoidal model with maximum 8, minimum -2, period 10, and maximum at $x = 3$.
10. Explain why $A\sin(B(x-C))+D$ is constant when $A = 0$, and why the usual fundamental-period formula does not apply.
11. A motor oscillates at 2.5 cycles per second. Determine its angular frequency and period.
12. Give the domain of $y = \tan(3x)$ and identify two consecutive vertical asymptotes.
13. Determine the amplitude and midline from a sinusoid with maximum 12.4 and minimum 7.6.

4 Inverse Trigonometry and Identities

Inverse trigonometric functions recover principal angles from trigonometric values. Their definitions require restrictions of the original domains. Trigonometric identities, in turn, express relationships among functions of the same angle and provide methods for simplifying and evaluating expressions.

Learning objectives. Determine the domains and ranges of the principal inverse trigonometric functions; evaluate and compose inverse and trigonometric functions; recover exact values from right triangles; prove identities using fundamental relations; and derive and apply angle-addition and double-angle formulas.

4.1 Inverse functions and principal branches

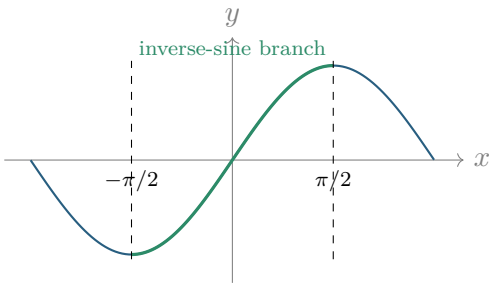
The notation $\sin^{-1} x$ commonly denotes $\arcsin x$, the inverse sine function. It must not be confused with $\csc x = 1/\sin x$. The inverse returns an angle from a prescribed trigonometric value, whereas the reciprocal returns a multiplicative inverse of that value.

A periodic function is not one-to-one on its full domain and therefore does not admit a global inverse function. To define the principal inverse sine, sine is restricted to $[-\pi/2, \pi/2]$, where it is strictly increasing and maps onto $[-1, 1]$. Similarly, cosine is restricted to $[0, \pi]$ and tangent to $(-\pi/2, \pi/2)$.

$f(x)$	Domain	Range
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

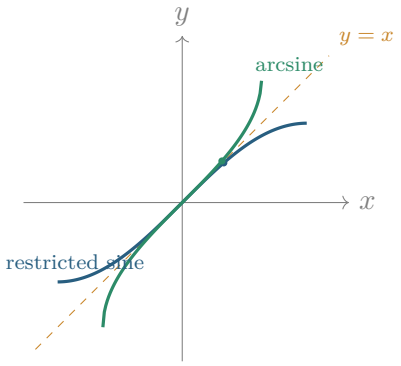
Example 4.1. $\arcsin(1/2) = \pi/6$, $\arccos(-1/2) = 2\pi/3$, and $\arctan(-1) = -\pi/4$.

Example 4.2. $\arcsin(\sin(5\pi/6)) = \pi/6$, not $5\pi/6$, because the inverse reports its principal value.



4.2 Reflection across the diagonal

The graphs of an invertible function and its inverse are reflections across the line $y = x$: if (a, b) lies on the graph of f, then (b, a) lies on the graph of f^{-1}. For inverse sine, only the restricted sine branch is reflected, rather than the entire periodic sine graph.



The reflected domains and ranges exchange roles. In particular, $\arcsin : [-1,1] \to [-\pi/2, \pi/2]$ whereas the restricted sine maps $[-\pi/2, \pi/2]$ onto $[-1,1]$.

Try it. List three points on $y = \cos x$ for $0 \le x \le \pi$, and interchange their coordinates to obtain points on $y = \arccos x$.

4.3 Compositions and triangle geometry

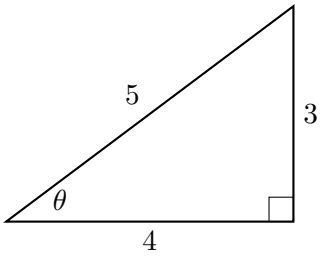
The identity $\sin(\arcsin x) = x$ holds for $-1 \le x \le 1$, but $\arcsin(\sin\theta) = \theta$ only for θ in the principal range. Suppose $\theta = \arctan(3/4)$. Then a right triangle has legs $3, 4$ and hypotenuse 5, so

$$\sin(\arctan(3/4)) = \frac{3}{5}.$$

For the inverse compositions, the domain restrictions determine which cancellations are valid:

$$\arcsin(\sin x) = x \quad \text{for } -\pi/2 \le x \le \pi/2, \qquad \sin(\arcsin u) = u \quad \text{for } -1 \le u \le 1.$$

The first identity does not hold for arbitrary real x, since the arcsine output is restricted to its principal range. Analogous statements hold for cosine and tangent on their chosen branches.



$\theta = \arctan(3/4)$

$\sin\theta = 3/5$

Example 4.3. To evaluate $\cos(\arcsin(-5/13))$, let $\theta = \arcsin(-5/13)$. Since $\theta \in [-\pi/2, \pi/2]$, its cosine is nonnegative. Thus

$$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}.$$

The principal range determines the sign of the square root.

Try it. Evaluate $\sin(\arccos(-3/5))$ exactly. Explain the sign of the result using the principal range of arccosine.

4.4 Identities and their domains

From the unit-circle equation we obtain

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

Dividing by $\cos^2 x$ (only where $\cos x \neq 0$) gives $1+\tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$ gives $1+\cot^2 x = \csc^2 x$ where $\sin x \neq 0$.

Example 4.4. For $\sin x \neq 0$,

$$\frac{1-\cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

This is an algebraic proof on the shared domain, not a numerical check at a few inputs.

4.5 Symmetry identities

The unit circle is symmetric about both coordinate axes. Reflection across the horizontal axis changes the sign of the vertical coordinate without changing the horizontal coordinate. Consequently,

$$\cos(-x) = \cos x, \qquad \sin(-x) = -\sin x.$$

The cosine function is *even*, while sine is *odd*. Taking quotients gives $\tan(-x) = -\tan x$ wherever tangent is defined.

A half-turn changes both coordinate signs:

$$\cos(x+\pi) = -\cos x, \qquad \sin(x+\pi) = -\sin x.$$

These relations explain the π-periodicity of tangent.

Try it. Use the unit-circle symmetries to evaluate $\sin(-7\pi/6)$ and $\cos(-7\pi/6)$ without a calculator.

4.6 Angle addition and double angles

Two successive plane rotations produce the sum identities

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Set $\beta = -\gamma$ for difference identities and $\beta = \alpha$ for double-angle identities:

$$\sin(2x) = 2\sin x\cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x.$$

Example 4.5. $\sin(75^\circ) = \sin(45^\circ + 30^\circ) = (\sqrt{6}+\sqrt{2})/4$.

4.7 A geometric derivation of the addition formulas

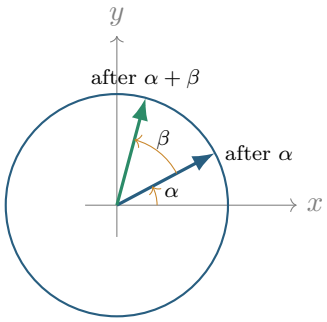
Rotation through an angle α sends a point (x, y) to

$$\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

In particular, apply this rotation to the unit-circle point $(\cos\beta, \sin\beta)$. The result must be the point $(\cos(\alpha+\beta), \sin(\alpha+\beta))$, since successive rotations add their angles. Matrix multiplication gives

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Thus the sum formulas express the geometry of composing rotations, rather than independent algebraic rules.



4.8 Double-angle and half-angle consequences

Setting $\alpha = \beta = x$ gives

$$\sin 2x = 2\sin x \cos x, \qquad \cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1.$$

Rearranging the cosine identity produces the power-reduction relations

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

These expressions are particularly useful when simplifying squared trigonometric functions.

Example 4.6. The exact value of $\cos 15^\circ$ follows by using $15^\circ = 45^\circ - 30^\circ$:

$$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Try it. Use the addition identities to find $\sin 105^\circ$ exactly. Then verify the sign from the quadrant containing 105°.

4.9 Applications: measured angles

An observation of a tower from 40 m away gives $\tan\theta = 3/4$. The elevation angle is $\arctan(3/4) \approx 36.9^\circ$, while the height is exactly $40(3/4) = 30$ m. Inverse functions recover angles; ratios can often recover distances directly.

4.10 Recovering angles in measurement problems

The inverse function identifies an angle from a ratio, provided the physical geometry determines an appropriate branch. If a right triangle has adjacent side 40 and opposite side 30, its acute angle satisfies

$$\theta = \arctan(30/40) = \arctan(3/4) \approx 36.87^\circ.$$

The same geometry gives the opposite side directly from $h = 40\tan\theta$, when the angle is already known. In a measurement context, the acute-angle restriction is supplied by the geometry of the triangle.

4.11 Exercises

1. Evaluate exactly $\arcsin(-\sqrt{2}/2)$ and $\arctan(\sqrt{3})$.

2. Explain why $\arcsin(\sin(3\pi/4)) \neq 3\pi/4$.

3. Find $\cos(\arcsin(5/13))$ with a triangle.
4. Find $\sin(\arctan(-2))$ exactly, including its sign.
5. Prove $(\sec^2 x - 1)/\tan x = \tan x$ on its domain.
6. Prove $(\sin x + \cos x)^2 = 1 + \sin(2x)$.
7. Calculate $\cos(75^\circ)$ exactly by angle addition.
8. Express $\cos(2x)$ using $\sin x$ alone.
9. State the domain and range of $\arccos x$, and explain why the restriction $[0, \pi]$ is appropriate.
10. Evaluate $\arccos(\cos(5\pi/3))$ exactly.
11. Find $\tan(\arccos(4/5))$ exactly using a right triangle.
12. Prove $\sin(-x) = -\sin x$ geometrically and use it to compute $\sin(-\pi/3)$.
13. Verify the cosine difference identity by replacing β with $-\beta$ in the cosine sum identity.
14. Prove $\cos^2 x = (1 + \cos 2x)/2$ on all of $\mathbb{R}$.
15. Find $\sin 15^\circ$ exactly and state its sign.
16. Explain why $\arctan(\tan(3\pi/4))$ is not $3\pi/4$.

5 Trigonometric Equations and Applications

Central question. When a trigonometric function has a prescribed value, how can we find *all* the angles that produce it? How can we use angles to solve triangles that are not right triangles?

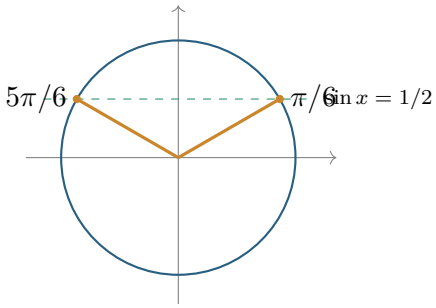
5.1 A first equation on the unit circle

Consider $\sin x = 1/2$. The unit circle has two points whose vertical coordinate is $1/2$, at $x = \pi/6$ and $x = 5\pi/6$. Every full revolution reproduces these points:

$$\boxed{x = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad x = \frac{5\pi}{6} + 2\pi k, \qquad k \in \mathbb{Z}.}$$

An inverse trigonometric function gives one principal angle, not necessarily every solution. The unit circle and periodicity give the complete answer.

Example 5.1. Solve $\cos x = -\sqrt{2}/2$ for $0 \le x < 2\pi$. The solutions are $3\pi/4$ and $5\pi/4$.



5.2 Equations with transformations

To solve $\sin(2x) = \sqrt{3}/2$ on $[0, 2\pi)$, first solve for $u = 2x$ on $[0, 4\pi)$. We obtain $u = \pi/3, 2\pi/3, 7\pi/3, 8\pi/3$ and therefore

$$x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}.$$

A changed frequency changes how many full cycles fit into the requested interval.

Example 5.2. Solve $2\cos x - 1 = 0$ on $[0, 2\pi)$. Isolating cosine gives $\cos x = 1/2$, hence $x = \pi/3$ or $5\pi/3$.

5.3 Factoring and checking domains

An equation may contain more than one trig function. Algebraic techniques still work:

$$2\sin^2 x - \sin x = 0 \quad \Longrightarrow \quad \sin x(2\sin x - 1) = 0.$$

On $[0, 2\pi)$ we obtain $x = 0, \pi, \pi/6, 5\pi/6$. Dividing by $\sin x$ at the outset would have lost two solutions.

5.4 Trigonometric identities in equations

For example, $2\cos^2 x - 1 = 0$ becomes $\cos^2 x = 1/2$, hence $\cos x = \pm\sqrt{2}/2$. There are four solutions on $[0, 2\pi)$. Using identities can make equations simpler, but each transformation should preserve the original domain.

5.5 Solving triangles: notation

Write the vertices of a triangle as A, B, C, and the opposite side lengths as a, b, c. Its angles satisfy $A + B + C = \pi$. For right triangles, the six trig ratios are enough. For oblique triangles, two laws extend our tools.

Definition 5.1 (Law of Sines)**.** For every nondegenerate triangle,

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.}$$

Example 5.3. A triangle has $A = 40°$, $B = 65°$, and $a = 9$ cm. Since $C = 75°$, the Law of Sines gives $b = 9 \sin 65° / \sin 40° \approx 12.69$ cm.

5.6 The ambiguous SSA case

Given two sides and a nonincluded angle (SSA), the Law of Sines may produce zero, one, or two triangles. If $\sin B = s$, both $B = \arcsin s$ and $B = \pi - \arcsin s$ must be considered, and each candidate must fit with the known angle A in a triangle.

Example 5.4. Suppose $A = 30°$, $a = 5$ and $b = 8$. Then $\sin B = 8 \sin 30°/5 = 0.8$. Both $B \approx 53.13°$ and $B \approx 126.87°$ are consistent with $A + B < 180°$, so two distinct triangles exist.

Definition 5.2 (Law of Cosines)**.** For a triangle with sides a, b, c opposite angles A, B, C,

$$\boxed{c^2 = a^2 + b^2 - 2ab \cos C.}$$

When $C = 90°$ this becomes the Pythagorean theorem. The cosine term measures how the included angle modifies the length of the opposite side.

Example 5.5. Two sides of a triangle are 7 m and 10 m, with included angle $60°$. The opposite side satisfies $c^2 = 49 + 100 - 140(1/2) = 79$, so $c = \sqrt{79}$ m.

5.7 Applied modeling and interpretation

Two sensors observe a distant object from different locations. The baseline and their sight angles make a triangle. The Law of Sines can recover an inaccessible distance if we know a side and two angles. Always include the diagram, angle units, and a reasonableness check.

5.8 Exercises

1. Solve $\sin x = -\sqrt{3}/2$ for $0 \le x < 2\pi$.
2. Solve $2 \sin(3x) = 1$ for $0 \le x < 2\pi$; find every solution.
3. Solve $2 \cos^2 x - \cos x = 0$ on $[0, 2\pi)$.
4. Solve $\tan x = -1$ for all real x.
5. A triangle has $A = 35°$, $B = 80°$ and $a = 12$. Find C and b.
6. For $A = 30°$, $a = 5$, $b = 8$, calculate both possible values of C in the SSA case.
7. Find the third side of a triangle when two sides have length 6 and 9 and their included angle is $120°$.

8. Explain why the Law of Cosines becomes the Pythagorean theorem for right triangles.

6 Vectors and Geometric Applications

Central question. How do we describe a quantity that has both a size and a direction?

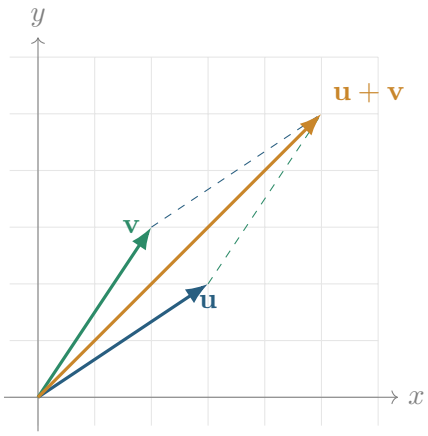
6.1 From coordinates to directed displacements

In the first half of the course we used $(\cos\theta, \sin\theta)$ to locate a point on a circle. A vector uses the same coordinates to describe a *displacement*: how far to move horizontally and vertically. Write $\mathbf{v} = \langle a, b\rangle$. Its magnitude is

$$\boxed{\|\mathbf{v}\| = \sqrt{a^2 + b^2}}.$$

Vectors with the same components represent the same displacement even when drawn at different starting points.

Example 6.1. A walker moves 3 km east and 4 km north. The resulting vector is $\langle 3, 4\rangle$ km and its magnitude is 5 km. The route traveled is 7 km, while the displacement is only 5 km.



6.2 Vector arithmetic

Addition combines displacements:

$$\langle a, b\rangle + \langle c, d\rangle = \langle a + c, b + d\rangle.$$

Scalar multiplication changes their length, and a negative scalar reverses direction:

$$k\langle a, b\rangle = \langle ka, kb\rangle.$$

Subtracting vectors compares locations. If $P = (p_1, p_2)$ and $Q = (q_1, q_2)$, then $\overrightarrow{PQ} = \langle q_1 - p_1, q_2 - p_2\rangle$.

Example 6.2. For $\mathbf{u} = \langle 2, -1\rangle$ and $\mathbf{v} = \langle -3, 4\rangle$, we have $\mathbf{u} + \mathbf{v} = \langle -1, 3\rangle$ and $2\mathbf{u} - \mathbf{v} = \langle 7, -6\rangle$.

6.3 Direction and trigonometry

A unit vector has magnitude 1. The unit vector pointing at angle θ is

$$\mathbf{e}_\theta = \langle \cos\theta, \sin\theta\rangle.$$

A vector of magnitude r in that direction is

$$\boxed{\mathbf{v} = r\langle \cos\theta, \sin\theta\rangle}.$$

Conversely, for a nonzero vector $\langle a, b\rangle$, the angle must be chosen in the appropriate quadrant. An inverse tangent applied only to b/a may miss that quadrant.

Example 6.3. A force of 20 N directed 30° above the horizontal has components $\langle 10\sqrt{3}, 10\rangle$ N.

6.4 The dot product measures alignment

Define

$$\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2.$$

The geometric interpretation is

$$\boxed{\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\phi},$$

where ϕ is the angle between nonzero vectors. Positive dot product means an acute angle; negative means an obtuse angle; zero means perpendicular vectors.

Example 6.4. With $\mathbf{u} = \langle 3, 4\rangle$ and $\mathbf{v} = \langle 4, -3\rangle$, the dot product is $12 - 12 = 0$. Both vectors are nonzero, so they are perpendicular.

6.5 Projection and useful components

The scalar component of $\mathbf{u}$ in the direction of a nonzero vector $\mathbf{v}$ is

$$\operatorname{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}.$$

The corresponding vector projection is

$$\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}.$$

Projection is the part of one vector aligned with another. In physics, the work done by a constant force $\mathbf{F}$ across displacement $\mathbf{d}$ is $W = \mathbf{F}\cdot\mathbf{d}$, measured in joules when force is in newtons and displacement in meters.

Example 6.5. A force $\langle 10, 5\rangle$ N acts during displacement $\langle 3, 0\rangle$ m. The work is 30 J: only the horizontal force component contributes.

6.6 A geometric connection to the Law of Cosines

For $\mathbf{u}, \mathbf{v}$ we have

$$\|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\phi.$$

Thus the Law of Cosines from Module 5 is also a theorem about vector differences.

6.7 Exercises

1. Find the vector from $P = (-2, 3)$ to $Q = (5, -1)$ and its magnitude.
2. For $\mathbf{u} = \langle 2, 5\rangle$, $\mathbf{v} = \langle -1, 3\rangle$, compute $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
3. Find exact components of a displacement of magnitude 12 at angle 150°.
4. Find a unit vector in the direction of $\langle -3, 4\rangle$.
5. Compute $\langle 2, -1\rangle \cdot \langle 5, 3\rangle$ and interpret its sign.
6. Determine whether $\langle 6, 2\rangle$ and $\langle -1, 3\rangle$ are perpendicular.

7. Find $\text{proj}_{\mathbf{v}}\,\mathbf{u}$ when $\mathbf{u} = \langle 4, 3\rangle$ and $\mathbf{v} = \langle 1, 0\rangle$.

8. A force of 50 N acts $60°$ from the direction of a 4 m displacement. Calculate the work done.

7 Polar Coordinates and Complex Numbers

Central question. What happens when we describe a point by its distance and direction instead of its horizontal and vertical coordinates?

7.1 Two descriptions of the same point

Cartesian coordinates specify (x, y). Polar coordinates specify (r, θ), where r is the directed radial distance and θ is an angle from the positive horizontal axis:

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta.}$$

For $r \geq 0$, the distance is $r = \sqrt{x^2 + y^2}$. The angle must be chosen using the signs of both coordinates, not just $\arctan(y/x)$.

Example 7.1. The polar point $(4, 5\pi/6)$ corresponds to $(x, y) = (-2\sqrt{3}, 2)$. Its distance from the origin is 4, and it lies in Quadrant II.

Try it. Convert $(r, \theta) = (6, -\pi/3)$ to Cartesian coordinates. Then describe its direction using an angle in $[0, 2\pi)$.

7.2 Polar coordinates are not unique

Angles that differ by 2π represent the same direction, so (r, θ) and $(r, \theta + 2\pi k)$ describe the same point. A negative radius reverses the ray:

$$(r, \theta) = (-r, \theta + \pi).$$

For example, $(2, \pi/3)$, $(2, 7\pi/3)$, and $(-2, 4\pi/3)$ are three descriptions of the same point. At the origin $r = 0$, every angle describes the same location.

7.3 Complex numbers as plane vectors

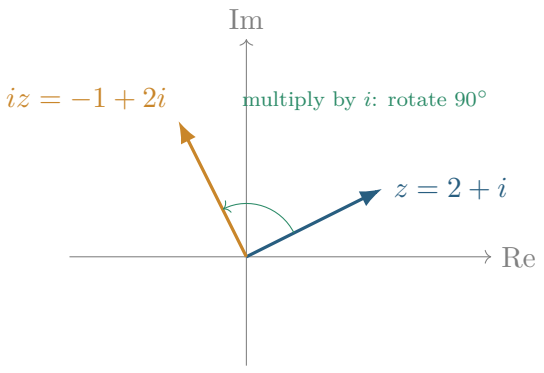
Introduce i satisfying $i^2 = -1$. A complex number $z = a + bi$ corresponds to the point (a, b) in the complex plane. Its modulus and argument are

$$|z| = \sqrt{a^2 + b^2}, \qquad \arg z = \theta \quad \text{when } z = |z|(\cos\theta + i\sin\theta).$$

The argument of $z = 0$ is undefined. As with polar angles, nonzero arguments are only unique after choosing a branch, such as $(-\pi, \pi]$.

Example 7.2. For $z = -\sqrt{3} + i$, the modulus is 2 and one argument is $5\pi/6$. Thus

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right).$$



7.4 Addition and multiplication mean different geometries

Complex addition follows the vector rule:

$$(a+bi)+(c+di)=(a+c)+(b+d)i.$$

Complex multiplication uses $i^2=-1$:

$$(a+bi)(c+di)=(ac-bd)+(ad+bc)i.$$

If $z=r(\cos\alpha+i\sin\alpha)$ and $w=s(\cos\beta+i\sin\beta)$, the angle-addition formulas give

$$\boxed{zw=rs(\cos(\alpha+\beta)+i\sin(\alpha+\beta)).}$$

Therefore multiplying complex numbers multiplies distances from the origin and *adds angles*. A unit complex number acts as a rotation. This unifies the unit circle from Module 1 with vector geometry from Module 6.

Example 7.3. Multiplying $2+i$ by i gives $i(2+i)=-1+2i$. The vector $(2,1)$ rotates counterclockwise by 90° to $(-1,2)$, without changing its length.

7.5 De Moivre's theorem and roots

For integer n, repeated multiplication gives

$$\boxed{[r(\cos\theta+i\sin\theta)]^n=r^n(\cos n\theta+i\sin n\theta)}$$

whenever the expression makes sense; for negative n, require $r>0$. For example, $(\cos(\pi/6)+i\sin(\pi/6))^3=i$. To solve $z^n=R(\cos\phi+i\sin\phi)$ with $R>0$, take

$$z_k=R^{1/n}\left(\cos\frac{\phi+2\pi k}{n}+i\sin\frac{\phi+2\pi k}{n}\right),\qquad k=0,\dots,n-1.$$

These n distinct roots are equally spaced around a circle.

Example 7.4. The four fourth roots of 1 are $1,i,-1,-i$. They are separated by a quarter-turn.

7.6 Polar graphs: a different way to draw a curve

In a polar equation such as $r=2\cos\theta$, the radius changes with the angle. Multiply by r: $r^2=2r\cos\theta$, or $x^2+y^2=2x$. Completing the square yields

$$(x-1)^2+y^2=1.$$

Thus a simple polar equation can represent an ordinary circle displaced from the origin.

7.7 Exercises

1. Convert $(5,2\pi/3)$ to Cartesian coordinates.
2. Give two distinct polar coordinate representations of $(\sqrt{2},\sqrt{2})$.
3. Write $-1-\sqrt{3}i$ in polar form with argument in $[0,2\pi)$.
4. Calculate $(2+i)(1-3i)$ in rectangular form.

5. Interpret multiplication by $-i$ as a plane transformation.
6. Evaluate $(\cos(\pi/4) + i\sin(\pi/4))^4$.
7. Find all three cube roots of 1 and plot them on the unit circle.
8. Convert the polar curve $r = 4\sin\theta$ into a Cartesian equation and identify the curve.

8 Parametric Equations and Conic Sections

Central question. How can the same curve be described by coordinates, by a moving point, or by a geometric rule?

8.1 A moving point traces a curve

In a parametric description, both coordinates depend on a parameter t:

$$\boxed{x = f(t), \qquad y = g(t).}$$

Think of t as time when a particle moves. The curve records its location, while the parameterization also records direction and timing.

Example 8.1. For $x = 2t + 1$, $y = t - 3$, eliminating $t = (x - 1)/2$ gives $y = (x - 7)/2$. If t increases, the point moves up and right along this line.

8.2 Eliminating the parameter

For $x = 3\cos t$, $y = 2\sin t$, square the normalized equations and add:

$$\boxed{\frac{x^2}{9} + \frac{y^2}{4} = 1.}$$

The equation gives the whole ellipse when t ranges over $[0, 2\pi]$. A restricted interval, such as $[0, \pi]$, traces only its upper half. Different parameterizations may have the same image but different speeds or directions.

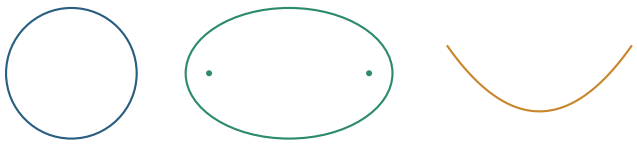
Example 8.2. Both $(x, y) = (\cos t, \sin t)$ and $(x, y) = (\cos 2t, \sin 2t)$ trace the unit circle for $0 \le t \le 2\pi$; the second makes two revolutions.

8.3 Projectile motion as a parametric curve

Under constant downward acceleration g, a projectile launched from (x_0, y_0) at speed v_0 and angle θ satisfies

$$x(t) = x_0 + v_0 \cos\theta\, t, \qquad y(t) = y_0 + v_0 \sin\theta\, t - \tfrac{1}{2} g t^2.$$

When the horizontal velocity is nonzero, eliminating t gives a parabola. The motion model adds information the Cartesian curve alone does not supply, including direction and location at a given time.



circle ellipse with foci parabola

8.4 Conics: circles and ellipses

A *circle* is the set of points at a fixed distance r from a center (h, k):

$$(x-h)^2 + (y-k)^2 = r^2.$$

An *ellipse* can be defined by a constant sum of distances to two foci. In standard form, for $a \geq b > 0$,

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1.$$

Its horizontal semiaxis is a and vertical semiaxis is b; when the major axis is horizontal, focal distance obeys $c^2 = a^2 - b^2$.

Example 8.3. For $x^2/25 + y^2/9 = 1$, the vertices are $(\pm 5, 0)$, and the foci are $(\pm 4, 0)$.

8.5 Parabolas and their focus

A *parabola* is the set of points equidistant from a fixed point (focus) and a fixed line (directrix). The equation

$$(x-h)^2 = 4p(y-k)$$

has vertex (h, k), focus $(h, k+p)$, and directrix $y = k - p$. Its axis is vertical; $p > 0$ opens upward and $p < 0$ downward.

Example 8.4. For $x^2 = 8y$, we have $4p = 8$, so the focus is $(0, 2)$ and the directrix is $y = -2$.

8.6 Hyperbolas and differences of distances

A *hyperbola* consists of points for which the absolute difference of distances to two foci is fixed. A horizontal hyperbola has standard form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

Its vertices are $(h \pm a, k)$, its foci are $(h \pm c, k)$ where $c^2 = a^2 + b^2$, and its asymptotes are $y - k = \pm (b/a)(x-h)$.

Example 8.5. The hyperbola $x^2/9 - y^2/16 = 1$ has vertices $(\pm 3, 0)$, foci $(\pm 5, 0)$, and asymptotes $y = \pm 4x/3$.

8.7 Changing descriptions

A parametric curve may be easier to generate than to recognize from a Cartesian equation. A Cartesian equation reveals symmetry or an axis. A focal definition explains why the curve arises in optics, navigation, and geometry. None of these descriptions replaces the others; each answers a different question.

8.8 Exercises

1. Eliminate t from $x = t + 2$, $y = 3t - 1$ and describe the motion as t increases.

2. Find the Cartesian equation traced by $x = 4\cos t$, $y = 3\sin t$.

3. Explain how $(\cos t, \sin t)$ and $(\cos 3t, \sin 3t)$ differ as parameterizations.

4. A projectile leaves the origin at 20 m/s at 30°, with $g = 9.8$ m/s^2. Write its parametric equations.
5. Identify center and radius of $(x-2)^2 + (y+1)^2 = 16$.
6. Find the foci of $x^2/36 + y^2/20 = 1$.
7. Find the focus and directrix of $(x-1)^2 = 12(y+2)$.
8. Find the vertices and asymptotes of $x^2/4 - y^2/9 = 1$.

9 Sequences and Series

Central question. How do repeated steps produce patterns, and how can we predict both a later value and a cumulative total?

9.1 A sequence is an ordered list

A sequence assigns a number a_n to each positive integer n. For example, $a_n = 3 + 2(n-1)$ produces $3, 5, 7, 9, \ldots$. We distinguish the *term* a_n from a *partial sum* $S_n = a_1 + \cdots + a_n$.

9.2 Arithmetic sequences: repeated addition

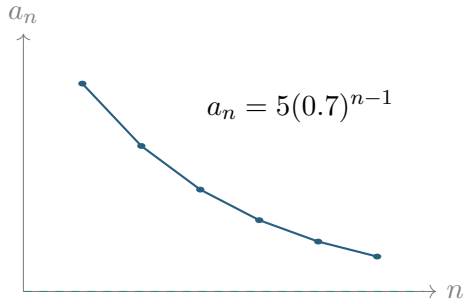
An arithmetic sequence has a fixed common difference d:

$$a_{n+1} = a_n + d, \qquad \boxed{a_n = a_1 + (n-1)d}.$$

Example 9.1. A theater's first row has 12 seats, with two more seats in each successive row. Row 15 has $12 + 14(2) = 40$ seats.

Adding the first and last terms in reverse order gives

$$\boxed{S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n-1)d)}.$$



9.3 Geometric sequences: repeated multiplication

A geometric sequence has a fixed common ratio q:

$$a_{n+1} = qa_n, \qquad \boxed{a_n = a_1 q^{n-1}}.$$

When $q > 1$, positive terms grow exponentially. When $0 < q < 1$, they decay toward zero.

Example 9.2. A population of 500 organisms increases by 8 percent each generation. The model is $a_n = 500(1.08)^{n-1}$ if generation 1 is the initial observation.

For $q \neq 1$, multiply the finite sum by q and subtract:

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1}, \qquad \boxed{S_n = a_1 \frac{1 - q^n}{1 - q}}.$$

If $q = 1$, the sum is na_1. The formulas refer to *finite* sums and work for ratios outside the interval $(0, 1)$ as well.

9.4 Sigma notation

The compact expression $\sum_{k=1}^{n} a_k$ means add the terms for integer indices $k = 1, \dots, n$. For instance,

$$\sum_{k=1}^{4} (2k+1) = 3 + 5 + 7 + 9 = 24.$$

The symbol $\sum$ tells us to add, while its bounds tell us which indices are included.

9.5 Infinite geometric series

If $|q| < 1$, then $q^n \to 0$. The finite geometric sum approaches

$$\boxed{\sum_{n=0}^{\infty} aq^n = \frac{a}{1-q}}, \qquad |q| < 1.$$

The condition matters: the series $1 + 2 + 4 + \cdots$ does not approach a finite sum.

Example 9.3. A ball bounces to 60% of its previous height. Released from height 5 m, the total vertical travel after infinitely many idealized bounces is $5+2(5)(0.6+0.6^2+\cdots) = 5+10(0.6/0.4) = 20$ m.

9.6 Recursive rules beyond arithmetic and geometric patterns

Not every sequence is arithmetic or geometric. Consider

$$a_{n+1} = \frac{a_n + 6}{2}, \qquad a_1 = 0.$$

The first terms are $0, 3, 4.5, 5.25, \dots$. Let $e_n = a_n - 6$ denote the difference from the fixed point. Then $e_{n+1} = e_n/2$, so $e_n = -6(1/2)^{n-1}$ and

$$a_n = 6 - 6(1/2)^{n-1}.$$

The original recursion is not geometric, but its *error from equilibrium* is. This is a useful bridge to feedback and dynamical models.

9.7 Exercises

1. Write the first five terms of $a_n = 7 - 3(n-1)$ and find a_{20}.
2. An arithmetic sequence has $a_1 = 4$, $d = 5$. Calculate S_{30}.
3. A geometric sequence has $a_1 = 81$, $q = 1/3$. Find a_6.
4. Compute $\sum_{k=0}^{5} 2(3/2)^k$ exactly.
5. Explain the difference between $\sum_{k=1}^{n} k$ and its final term n.
6. Determine whether $5 + 5/2 + 5/4 + \cdots$ converges and find its sum.
7. A machine loses 12 percent of its value per year. Express its value after n years if it initially costs \$12,000.
8. For $a_{n+1} = (a_n + 10)/2$, $a_1 = 2$, find the fixed point and an explicit formula for a_n.

10 Synthesis: Motion and Patterns

Central question. How can the different mathematical languages of this course describe one evolving system?

10.1 One motion, several descriptions

Imagine a point traveling counterclockwise around a circle of radius R with angular speed ω. Its angle at time t is $\theta(t) = \theta_0 + \omega t$. In parametric form,

$$x(t) = R\cos(\theta_0 + \omega t), \qquad y(t) = R\sin(\theta_0 + \omega t).$$

Its polar form is $(r, \theta) = (R, \theta_0 + \omega t)$, and its complex form is

$$\boxed{z(t) = R(\cos(\theta_0 + \omega t) + i\sin(\theta_0 + \omega t))}.$$

These are not competing models: they encode the same physical position in different coordinate languages.

10.2 Velocity, vectors, and periodicity

Over a short time interval Δt, the displacement vector is

$$\Delta\mathbf{r} = \langle x(t + \Delta t) - x(t), y(t + \Delta t) - y(t)\rangle.$$

For constant angular speed the point completes a revolution after $T = 2\pi/|\omega|$ when $\omega \neq 0$. Its horizontal and vertical coordinates repeat with that period. The direction of motion is tangent to the circle and perpendicular to the radius: this is the geometric heart of circular motion.

10.3 Sampling creates a sequence

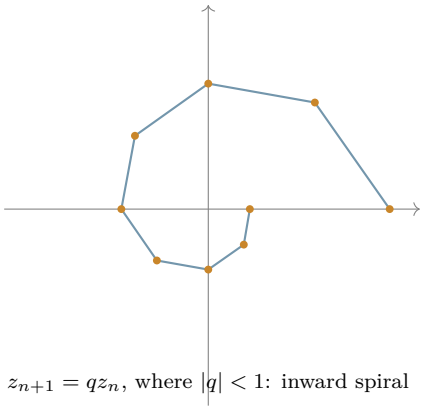
Suppose we record the position every Δt seconds. The resulting sequence of complex positions is

$$z_n = R(\cos(\theta_0 + n\delta) + i\sin(\theta_0 + n\delta)), \qquad \delta = \omega\Delta t.$$

Writing $u = \cos\delta + i\sin\delta$ gives

$$\boxed{z_{n+1} = uz_n, \qquad z_n = u^n z_0.}$$

De Moivre's theorem then explains every sample as a repeated rotation. If $\delta = 2\pi/m$ for a positive integer m, the sampled orbit closes after m steps (provided this is the smallest such positive step count). If $\delta/(2\pi)$ is irrational, no two different samples have exactly the same position.



$z_{n+1} = qz_n$, where $|q| < 1$: inward spiral

10.4 Scaling and rotation together

Now consider the recurrence $z_{n+1} = qz_n$ where

$$q = \rho(\cos\delta + i\sin\delta).$$

Its explicit solution is $z_n = q^n z_0$. Consequently,

$$|z_n| = \rho^n |z_0|, \qquad \arg z_n \equiv \arg z_0 + n\delta \pmod{2\pi}$$

for nonzero terms. If $0 < \rho < 1$, the points spiral toward the origin; if $\rho > 1$, they spiral outward; and if $\rho = 1$, they remain on a circle. Thus a geometric sequence and a rotation combine to produce a spiral.

10.5 Sums represent accumulated effects

If a system moves by step vectors $\mathbf{v}_0, \mathbf{v}_1, \ldots$, its total displacement after N steps is the vector sum $\sum_{n=0}^{N-1} \mathbf{v}_n$. For a complex-valued geometric pattern $\Delta z_n = aq^n$ with $q \neq 1$,

$$z_N - z_0 = \sum_{n=0}^{N-1} aq^n = a\frac{1-q^N}{1-q}.$$

Notice that the location sequence z_n and the step sequence Δz_n are different objects. Careful modeling starts by distinguishing what is being measured.

10.6 Different curves arise from different constraints

Circular and elliptical motion can be expressed parametrically. A projectile follows a parabola when horizontal velocity is constant and vertical acceleration is constant. Polar formulas express rays and radial patterns directly. The useful representation depends on whether we need a shape, a position, a direction, or a rule for change.

10.7 A complete example

A sensor begins at $z_0 = 2$ in the complex plane. Every second its controller multiplies the position by $q = \frac{1}{2}(\cos 60^\circ + i\sin 60^\circ)$. After n steps,

$$z_n = 2\left(\frac{1}{2}\right)^n (\cos(60^\circ n) + i\sin(60^\circ n)).$$

After three steps the modulus is $1/4$, and the direction is 180°. Hence $z_3 = -1/4$. The modulus obeys a geometric sequence, the direction obeys an arithmetic progression of angles, and the position follows a discrete inward spiral.

10.8 Exercises

1. Write Cartesian, polar, and complex formulas for a circle of radius 5 rotating once every 20 seconds, starting on the positive horizontal axis.

2. Sample that motion every five seconds. Give its first five complex positions and identify its repetition period in samples.

3. For $z_{n+1} = iz_n$ and $z_0 = 1 + i$, calculate $z_1, \ldots, z_4$ and describe the geometry.

4. For $z_{n+1} = (1/2)iz_n$, $z_0 = 4$, find z_4 and explain the change in magnitude and argument.

5. A point travels along $x = 3\cos t$, $y = 2\sin t$. Eliminate the parameter and describe one complete traversal.

6. Explain why a constant-speed circular trajectory is periodic but the spiral $z_n = (0.8e^{i\pi/3})^n z_0$ is not periodic when $z_0 \neq 0$.

7. Compute the finite sum $1 + i + i^2 + i^3$ and interpret its geometric cancellation.

8. Describe a real-world process that could be analyzed with at least three of the course's mathematical representations.